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1 GLM Overview

1.1 Example

Normal Regression: $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$: $\epsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$

Logistic Regression: $y_i \overset{ind}{\sim} Bern(p_i)$: $logit(p_i) = \beta_0 + \beta_1 x_i$, $log(\frac{p_i}{1-p_i})$

Differences:

- Support of y_i: $y_i \in \mathbb{R}$ vs. $y_i \in \{0, 1\}$
- Link functions: $g(\mu_i) = \mu_i$ vs. $g(p_i) = log(\frac{p_i}{1-p_i})$
- Likelihoods: $y_i \overset{ind}{\sim} N(\mu_i, \sigma^2)$ vs. $y_i \overset{ind}{\sim} Bern(p_i)$

Similarities:

- Univariate models
- Exponential families
- Indepdently distributed
- Linear (or otherwise systematic) relationship with a function of $E(y_i)$
- Have link functions
- Assuming that y_i is random
- β's are unknown

1.2 GLMs

Generalized Linear Model:

- Random component: $Y_i \sim f(\mu_i)$
- Link component: $g(\mu_i) = \eta_i$
- Systematic component: $\eta_i = x_i^\top \underset{\sim}{\boldsymbol{\beta}}$

We generally use LMs for the following:

- Inference: understand relationships
- Prediction: hevaior of specific x's

1.3 Poisson Regression

$Y_i \overset{\text{ind}}{\sim} Poisson(\mu_i), log(\mu_i) = x_i^\top \beta$

Assumptions:

- Linearity
- Independence

- Poisson distribution on response var

1.4 Exponential Family Calculations

With the Exponential Family log-likelihood written as: $l(\theta; y) = w(\theta)t(y) + ln(a(\theta)) + ln(b(y))$ where $ln(a(\theta)) = c(\theta)$ and $ln(b(y)) = d(y)$:

$Score = w'(\theta)t(y) + c'(\theta)$

$\mathbb{E}[t(y)] = -\frac{c'(\theta)}{w'(\theta)}$

$Var(t(y)) = \frac{w''(\theta)c'(\theta) - w'(\theta)c''(\theta)}{(w'(\theta)^3)}$

Example: Poisson: $f(y;\lambda) = \frac{\lambda^y e^{-\lambda}}{y!} = exp\{yln(\lambda - \lambda - ln(y!))\}$

In this case, $c(\lambda) = -\lambda$ and $w(\lambda) = ln(\lambda)$

Sufficient Statistic: $t(y) = y$

$w'(\lambda) = \frac{1}{\lambda}, w''(\lambda) = -\frac{1}{\lambda^2}, c'(\lambda) = -1, c''(\lambda) = 0$

Mean: $\mathbb{E}[t(y)] = \mathbb{E}[y] = -\frac{c'(\lambda)}{w'(\lambda)} = -\frac{1}{\frac{-1}{\lambda}} = \lambda$

Variance: $Var(t(y)) = Var(y) = \frac{w''(\lambda)c'(\lambda) - w'(\lambda)c''(\lambda)}{(w'(\lambda))^3} = \frac{(-1)(-\frac{1}{\lambda^2}) - 0}{\frac{1}{\lambda^3}} = \frac{\frac{1}{\lambda^2}}{\frac{1}{\lambda^3}} = \lambda$

Score: $w'(\theta)t(y) + c'(\theta) = (\frac{1}{\lambda})y + (-1)$. Verify $\mathbb{E}[Score] = 0 : \mathbb{E}[\frac{1}{\lambda}y - 1] = \frac{1}{\lambda}\mathbb{E}[y] - 1 = \frac{\lambda}{\lambda} - 1 = 0$

1.5 Link Function

$\eta_i = g(\mu_i)$

Requirements (all with respect to the mean μ, not x):

- One-to-One ($g(a) = g(b) \implies a = b$)
- Monotone (Monotone increasing means $a > b \implies g(a) > g(b)$)
- Continuous and differentiable ($lim_{\mu \to c} g(\mu) = g(c)$)

Canonical Link: $g * (w(\theta_i)) = w(\theta_i)$

Example,

Binomial: $w(\pi_i) = log(\frac{\pi_i}{1-\pi_i}) \implies g(\pi_i) = log(\frac{\pi_i}{1-\pi_i}), g * (w(\pi_i)) = w(\pi_i)$

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With U as the score function, and $\mathbb{J}$ as the variance:

$U \sim N(\underset{\sim}{\mathbf{0}}, \mathbb{J})$

$\mathbb{J}^{-1/2}U \sim N(0, 1)$

$(\mathbb{J}^{-1/2}U)^\top \mathbb{J}^{-1/2}U = U^\top \mathbb{J}^{-1}U \sim \chi^2(k)$

3 Prediction Intervals

Using regular process for confint on predictions of using the variance of $\frac{X\mathbb{I}X^\top}{g^{-1}(\hat{\mu})}$ provides too large of an interval. The better method is to get interval on $\hat{\eta}$ then transform bounds to $\hat{\mu}$:

$\hat{\eta} = x\hat{\beta} \sim N(x\beta, XCov(\hat{\beta})X^\top)$ Use that variance to get $SE(\hat{\eta})$

$\hat{\eta} \pm 1.96SE(\hat{\eta}) = [\hat{\eta}_1, \hat{\eta}_2], \hat{\mu}_2 = g^{-1}(\hat{\eta}_2)$

4 Residuals

Perason Residuals: $r_{p_i} = \frac{y_i - \hat{\mu}_i}{\sqrt{Var(Y_i)}}$

Examples:

Poisson: $\frac{y_i - \hat{\mu}_i}{\sqrt{\hat{\mu}_i}}$

Logit: $\underline{y_i - \hat{\mu}_i}$

Anscombe: R doesn't really even touch

Deviance: uses the individual observation log-likelihood

Working: Similar to the Z statistic (difference from mean divided by SE)

5 Picking Link Functions

There are lots of ways to pick link functions:

- Interpretation
- AIC/BIC
- Specificity
- etc.

In some cases, the link function doesn't make much of a difference. However, sometimes it does.

6 Crabs Analysis

Model with 'is.very.dark' as the covariate. 2 common ways of writing:

1.

$\eta_{ij} = \alpha_0 + \gamma_j$: Y_{ij} = whether or not the ith crab in the jth color group has a sateliitle ($j = 1, 2; i = 1, ..., n$)

Often constrained such that $\eta_{i1} = \alpha_0 + \gamma_1, \eta_{i2} = \alpha_0 + \gamma_2, \gamma_1 + \gamma_2 = 0$ ("sum to zero" constraint)

ie. $x = \begin{pmatrix} 1 & -1 \\ 1 & -1 \\ 1 & 1 \\ 1 & 1 \\ \vdots & \end{pmatrix}$

2.

$\eta_i = \beta_0 + I(x_i = True)\beta_1$

When written with the indicator, that's the same as constraint $\gamma_1 = 0$

ie. $x = \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 1 \\ 1 & 1 \\ \vdots & \end{pmatrix}$

Note:

$\hat{\beta}_0 = \hat{\alpha}_0 + \hat{\gamma}_1$

$\hat{\beta}_0 + \hat{\beta}_1 = \hat{\alpha}_0 + \hat{\gamma}_2$

β_0 is the log odds that a not very dark crab has a satellite

e^{β_0} is the odds that a not very dark crab has a satellite

$\beta_0 + \beta_1$ is the log odds that a very dark crab has a satellite

$e^{\beta_0+\beta_1}$ is the odds that a very dark crab has a satellite

π_{i1} is the probability of satellite for not very dark

π_{i2} is the probability of satellite for very dark

$ln(\frac{\pi_{i2}}{(1-\pi_{i2})}) = \beta_0 + \beta_1$

$ln(\frac{\pi_{i1}}{(1-\pi_{i1})}) = \beta_0$

$ln(\frac{\pi_{i2}/(1-\pi_{i2})}{\pi_{i1}/(1-\pi_{i1})}) = \beta_1$ which is the log odds ratio that a crab has a satellite for a very dark vs. not very dark

Now with a 4-level factor as the covariate

$\eta_i = \beta_0 + I(x_i = "DarkMed")\beta_{dm} + I(x_i = "LightMed)\beta_{lm} + I(x_i = "Med")\beta_{med}$

$H_0 : \beta_{dm} = \beta_{lm} = \beta_{med} = 0, H_1 :$ at leat one is not 0

Test with Delta Deviance and compare to $\chi^2(3)$. We get a p-value of .0033 so we rejec the H_0 and conclude there is enough evidence that at least of the βs should be in the model.

For testing a specific β:

$H_0 : \beta_{dm} = 0, H_1 : \beta_{dm} \neq 0$. p-value is 0.04. Any significance less than this would find that dm is not different from dark

Now, for testing contrasts between βs:

$H_0 : \beta_{med} - \beta_{lm} = 0, H_1 : \beta_{med} - \beta_{lm} \neq 0$

Contrast: $c = (0, 0, -1, 1)^\top$, usual contrast rules apply

$e^{\beta_{med}}$ is the odds ratio that a crab has a satellite for mdeium vs. dark crabs $(\eta_{i,med} - \eta_{i,d} = \beta_{med}, log(\frac{\pi_{1,med}/(1-\pi_{i,med})}{\pi_{1,d}/(1-\pi_{i,d})}))$

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Final model from selection:

$y_i \overset{\text{ind}}{\sim} Bern(\pi_i)$

$ln(\frac{\pi_i}{1-\pi_i}) = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}$

x_{i1} represents the width of crab i, x_{12} represents the indicator for 'is very dark' for crab i

For a 1 unit increase in width, the log-odds increase by β_1.

The odds change by a factor of e^{β_1} when going from one width to a width of 1 unit more.

e^{β_1} is the odds-ratio of having a satellite for two crabs one unit width difference. "Reversing" it, e^{β_1} is the odds-ratio of a crab being one unit wider for a crab that does or does not have a satellie.